Lect No.(6-7)

HTML: Block & Inline Elements, Attributes, and Layouts

HTML elements can be broadly categorized into **Block-level** and **Inline-level** elements. These, along with **attributes, headings, paragraphs, tables, lists, and layout tags**, are fundamental for structuring web pages.

Block and Inline Elements

Block-level Elements

- **Characteristics:**
  - Take up the **entire width** of the parent container.
  - Always start on a **new line**.
  - Can contain **other block-level and inline elements**.
- **Common Examples:**

```
<div>, <p>, <h1> ... <h6>, <section>, <article>, <header>, <footer>,
<aside>, <main>.
```

- **Use Cases:**
  - Structuring **sections** of a page.
  - Wrapping multiple elements together.
  - Creating **containers** for styling via CSS.

Code Example:

```
<div>
   <h1>Main Heading</h1>
   <p>This paragraph is block-level and starts on a new line.</p>
</div>
```

Inline Elements

- **Characteristics:**
  - Take only the **width required** by the content.
  - Do **not** start on a new line.
  - Usually used for text formatting or small pieces of content.
- **Common Examples:**

```
<span>, <a>, <img>, <strong>, <em>, <mark>, <code>.
```

- **Use Cases:**
  - Highlighting text.
  - Embedding links or images within text.
  - Styling specific words or phrases.

Code Example:

```html
<p>This is <span style="color:red;">inline text</span> inside a paragraph.</p>
```

Comparison Table

Feature	Block Element	Inline Element
Starts on new line?	Yes	No
Width	Full width of container	Only as wide as content
Can contain	Block & inline	Only inline
Examples	<div>, <p>, <section>	<span>, <a>, <strong>

Attributes

Attributes **provide additional information** about HTML elements and modify behavior or appearance.

Common Attributes and Use Cases:

Attribute	Description	Example
id	Unique identifier for an element	<p id="para1">Hello</p>
class	CSS class for styling multiple elements	<div class="container"> Content </div>
style	Inline CSS styles	Blue text
src	Source for images, scripts	<img src="pic.jpg" alt="Picture">
href	URL for links	Google
alt	Alternative text for images	<img src="cat.jpg" alt="Cat">
title	Tooltip text	<p title="Hover me">Text</p>
value	Input value in forms	<input value="Hello">
selected	Pre-selected dropdown option	<option selected>Apple</option>

disabled	Disables a form element	`<input disabled>`
placeholder	Placeholder text for inputs	`<input placeholder="Enter name">`

Example:

```
<input type="text" id="username" class="input-field" placeholder="Enter
Name">
```

Headings

- HTML supports **6 levels of headings** `<h1>` to `<h6>`.
- Use **hierarchically** for SEO and accessibility.
- **Block-level by default**.

Example:

```
<h1>Main Heading</h1>
<h2>Subheading</h2>
<h3>Section Heading</h3>
<h4>Subsection</h4>
<h5>Smaller Heading</h5>
<h6>Minor Heading</h6>
```

Best Practices:

- Only **one `<h1>` per page** (main topic).
- Use headings to create **logical page structure**.

Paragraphs

- `<p>` defines a **paragraph of text**.
- Can contain **inline elements** for formatting.

Example:

```
<p>This is a <strong>paragraph</strong> with <em>emphasized</em> and
<mark>highlighted</mark> text.</p>
```

Attributes:

- `align` → Align text (`left`, `center`, `right`) *(deprecated, use CSS instead)*

Tables

HTML tables are used to **display structured data in rows and columns**. They are widely used for reports, schedules, comparison charts, etc.

Basic Structure

Tag	Purpose
`<table>`	Container for the table
`<caption>`	Table title or description
`<tr>`	Table row

`<th>`	Table header (bold & centered by default)
`<td>`	Table cell (data)
`<thead>`	Groups header rows
`<tbody>`	Groups body rows
`<tfoot>`	Groups footer rows

Example:

```
<table border="1">
  <caption>Student Marks</caption>
  <thead>
    <tr>
      <th>Name</th>
      <th>Math</th>
      <th>Science</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>John</td>
      <td>95</td>
      <td>88</td>
    </tr>
    <tr>
      <td>Jane</td>
      <td>89</td>
      <td>92</td>
    </tr>
  </tbody>
```

```
  <tfoot>
    <tr>
      <td>Total</td>
      <td>184</td>
      <td>180</td>
    </tr>
  </tfoot>
</table>
```

Important Table Attributes

Attribute	Purpose	Example
`border`	Border thickness	`<table border="2">`
`cellpadding`	Space inside cells	`<table cellpadding="10">`
`cellspacing`	Space between cells	`<table cellspacing="5">`
`colspan`	Merge columns	`<td colspan="2">Merged</td>`
`rowspan`	Merge rows	`<td rowspan="2">Merged</td>`
`align`	Alignment of cell (deprecated)	`<td align="center">Text</td>`

Nested Tables

Tables can be nested for **complex layouts**:

```
<table border="1">
  <tr>
    <td>
      <table border="1">
        <tr><td>Nested 1</td></tr>
        <tr><td>Nested 2</td></tr>
      </table>
    </td>
    <td>Main Table Cell</td>
  </tr>
</table>
```

Best Practices for Tables

- Use `<thead>`, `<tbody>`, `<tfoot>` for **readability and accessibility**.
- Avoid using tables for **layout purposes**; use CSS instead.
- Always provide **captions** to describe the table.
- Use **semantic** **`<th>`** for headers instead of styling `<td>` bold.

Dropdowns (Select Menu)

- Created with `<select>` and `<option>` elements.
- Allows **users to select one or multiple options**.

Example:

```
<label for="fruits">Choose a fruit:</label>
<select id="fruits" name="fruits">
  <option value="apple">Apple</option>
  <option value="banana">Banana</option>
  <option value="mango" selected>Mango</option>
</select>
```

Attributes:

- `multiple` → Allow multiple selections
- `disabled` → Disable dropdown

Quotations

Inline Quotes → `<q>`

- Automatically adds **quotation marks**.

```
<p>He said, <q>HTML is fun!</q></p>
```

Block Quotes → `<blockquote>`

- Used for **long quotations**.
- Optional `cite` attribute for source URL.

```
<blockquote cite="https://www.example.com">
  HTML is the backbone of every website.
</blockquote>
```

Lists

HTML lists are used to **group related items** in an organized way. Lists can be **ordered, unordered, or description lists**.

Unordered List `<ul>`

- Used for items with **no specific order**.
- Items are wrapped in `<li>` tags.
- Can be **styled with bullets, squares, circles** via CSS.

Example:

```html
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Attributes:

- `type` → Bullet style (disc, circle, square) *(deprecated, use CSS instead)*

Ordered List `<ol>`

- Used for **numbered or sequential items**.
- Items wrapped in `<li>`.

Example:

```html
<ol>
  <li>Install VS Code</li>
  <li>Create project folder</li>
  <li>Run live server</li>
</ol>
```

Attributes:

- `type` → Numbering style (1, a, A, i, I)
- `start` → Starting number
- `reversed` → Reversed order

Example with Attributes:

```html
<ol type="A" start="3" reversed>
  <li>Step C</li>
  <li>Step B</li>
  <li>Step A</li>
</ol>
```

Description List <dl>

- Used for **term-description pairs**.
- Tags: `<dl>` (list), `<dt>` (term), `<dd>` (description).

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Nested Lists

- Lists can be **nested inside another list** to create hierarchy.

Example:

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Mango</li>
    </ul>
  </li>
  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Spinach</li>
    </ul>
  </li>
</ul>
```

Lists in Forms and Menus

- Dropdowns often **use `<ul>` for custom menus** in navigation.
- Nested lists are useful for **multi-level menus**.

Example:

```
<nav>
  <ul>
    <li>Home</li>
    <li>About</li>
    <li>Services
      <ul>
        <li>Web Development</li>
        <li>App Development</li>
      </ul>
    </li>
    <li>Contact</li>
  </ul>
</nav>
```

Best Practices for Lists

- Use `<ul>` for **unordered**, `<ol>` for **ordered**, `<dl>` for **definitions**.
- Always wrap **list items with `<li>`**.
- Avoid excessive nesting for readability.
- Use **CSS** to style bullets, numbers, and indentation.
- Nested lists are ideal for **navigation menus and subtopics**.

Blocks

Generic container elements used to structure content.

Common Elements:

Element	Purpose
`<div>`	Generic block container
`<section>`	Group related content
`<article>`	Independent content (blog, news)
`<header>`	Top section of page
`<footer>`	Bottom section
`<aside>`	Sidebar content

Example:

```
<section>
  <h2>About Me</h2>
  <p>I am a web developer specializing in frontend development.</p>
</section>
```

Layout Elements

- **Semantic elements** improve **readability, accessibility, and SEO**.

Common Layout Elements:

- `<header>` → Page header with logo/navigation
- `<nav>` → Navigation menu
- `<main>` → Main page content
- `<section>` → Section of content
- `<article>` → Independent content block
- `<aside>` → Sidebar or supplementary content
- `<footer>` → Page footer with copyright/contact

Example:

```
<body>
  <header>
    <h1>Website Header</h1>
    <nav>
      <a href="#">Home</a> | <a href="#">About</a>
    </nav>
  </header>

  <main>
    <article>
      <h2>Blog Post</h2>
      <p>Content of the blog post goes here.</p>
    </article>
    <aside>
      <h3>Related Links</h3>
      <p>See more tutorials.</p>
    </aside>
  </main>

  <footer>
    <p>(c) 2025 My Website</p>
  </footer>
</body>
```